

Continuous Distillation

Final Lab Report
Unit Operations Lab 6
April 21st, 2026

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Executive Summary

The objective of this experiment is to evaluate the performance of an operating continuous distillation column in separating an ethanol-water mixture and to determine the influence of the location of feed on the separation efficiency of the column in design and operational considerations.

A steady state continuous distillation column was run with a binary ethanol-water feed. A reboiler was used to provide heat to make vapor, and a condenser was used to provide reflux to increase separation. The feed was added at various positions in the column and samples of the distillate stream, and the bottoms stream was taken. Densitometers were used to determine product compositions, and temperature profiles were plotted along the column. Vapor liquid equilibrium relationships and McCabeThiele methodology were used to analyze the experimental data. In the case of middle feed conditions (Experiment B), the distillate and bottoms stream were found to contain an average composition of 89.99 wt% ethanol (73.50 mol%), and 12.52 wt% ethanol (4.24 mol%), respectively. In the case of the best feed condition (Experiment C), the distillate composition reduced to 88.46 wt% ethanol (70.30 mol%), and the bottoms stream enriched to 16.20 wt% ethanol (5.63 mol%). These findings show that there was a decline in separation performance when the feed was added at some other point than the ideal point.

The results indicate that the column achieved effective separation under both conditions; however, optimal feed placement is critical for maximizing efficiency. Deviations of ideal behavior were monitored due to the non-equilibrium stage operation, heat loss, and variability. Overall, the results of the experiment agree with distillation theory and demonstrate that separation efficiency is decreased when the location of the feed is not appropriate.

2. Introduction

The separation of liquid mixtures is a fundamental operation in the chemical process industries, with distillation being one of the most widely used techniques. Continuous distillation is particularly important in large-scale operations due to its ability to provide consistent product quality and efficient separation. In this experiment, a binary ethanol–water system was separated using a continuous distillation column to evaluate column performance and understand the impact of operating conditions on separation efficiency.

Distillation operates on the principle of differences in component volatility, where the more volatile component preferentially enters the vapor phase while the less volatile component remains in the liquid phase. Through repeated vapor–liquid contact within the column, the vapor becomes enriched in ethanol as it rises, while the liquid becomes enriched in water as it descends. The effectiveness of this separation depends on several factors, including reflux ratio, feed condition, and feed location within the column. The objective of this experiment was to evaluate the performance of a continuous distillation column and assess the effect of feed location on separation efficiency. The column was operated at a steady state, and the feed was introduced at different tray locations to observe changes in product composition. Distillate and bottoms samples were collected and analyzed using a densitometer to determine ethanol concentration, while temperature

measurements were recorded along the column to assess internal conditions. Understanding the relationship between operating conditions and separation performance is essential for the design and optimization of industrial distillation systems. The results from this experiment provide practical insight into how deviations from ideal operation, such as non-optimal feed placement, can impact column efficiency and product purity.

3. Theory (10 pts)

Distillation is an equilibrium-based separation process that exploits differences in component volatility to achieve separation of liquid mixtures. In this experiment, a binary ethanol-water system was separated in a continuous distillation column operating at steady state. The feed was introduced near the middle of the column, while heat supplied in the reboiler generated vapor that rose counter currently to the descending liquid. Ideally, each stage in the column approached vapor-liquid equilibrium (VLE), allowing progressive enrichment of the more volatile component (ethanol) in the vapor phase and the less volatile component (water) in the liquid phase (UIC, 2024).

The basis for analyzing binary distillation columns is the McCabe-Thiele method, which provides a graphical solution for determining the number of theoretical stages required for a given separation. This method relies on equilibrium data expressed as a relationship between vapor composition y and liquid composition x , typically obtained from vapor-liquid equilibrium data for the ethanol-water system (Wankat, 2012). The operating lines for the rectifying and stripping sections of the column are derived from component mass balances and depend on the reflux ratio and feed condition:

For a saturated liquid feed, the feed line (q-line) has a slope of:

The feed line (q-line) equation for a saturated liquid feed is:

The rectifying operating line is given by:

Where R is the reflux ratio and y_D is the distillate composition. The stripping line is similarly derived from mass balances below the feed stage:

The number of theoretical stages is determined by stepping between the equilibrium curve and operation lines on an x - y diagram (McCabe, et al., 2005).

In practice, the assumption of equilibrium on each tray is not fully realized. To account for this, the concept of Murphree tray efficiency is introduced. This efficiency quantifies how closely an actual stage approaches equilibrium and is defined as:

Where y_n is the vapor composition in equilibrium with the liquid on stage n (UIC, 2024). Lower efficiencies result in a greater number of actual stages being required to achieve the same separation predicted by the ideal McCabe-Thiele method.

The overall column efficiency is determined by:

The ethanol-water system exhibited non-ideal behavior due to molecular interactions, and its VLE data must be obtained experimentally or from literature correlations. This data is typically presented as y versus x curves at a fixed pressure, which are used directly in the McCabe-Thiele construction. Deviations from ideality can be described using activity coefficient models such as Wilson or NRTL, although for this experiment, tabulated equilibrium data was sufficient (ChemLibreTexts, 2023).

The experimental data collected in this experiment included temperature profiles (T1-T9) along the column and compositions of the distillate and bottoms streams, determined using a densitometer. Since tray temperatures are related to composition through VLE relationships, the temperature profile provided qualitative insight into the internal composition gradient of the column. At steady state, a stable temperature profile indicated consistent separation performance. The measured product compositions were used to construct operating lines and perform a McCabe-Thiele analysis, enabling comparison between theoretical predictions and actual column performance (UIC, 2024).

To evaluate the column performance quantitatively, the experimentally determined number of stages was compared to the theoretical number of stages predicted by the McCabe-Thiele method. The ratio of these values provided an estimate of the overall column efficiency. Additionally, an energy balance around the column may be used to assess heat losses and deviations from ideal assumptions, as real systems often experiment heat transfer to the surroundings and incomplete phase equilibrium (Seader et al., 2016).

The effect of feed location was a key variable in this experiment. When the feed was introduced at the optimal location, the separation efficiency was maximized. Deviating from this location, such as introducing the feed above the top tray, alters the internal vapor-liquid traffic and reduces separation efficiency. This behavior is predicted by

distillation theory and can be observed experimentally through changes in product composition and temperature profiles (Towler & Sinnott, 2013).

Overall, the experimental results were analyzed by comparing measured compositions and temperature data with the theoretical predictions from the McCabe-Thiele method. Discrepancies between theory and experiment were attributed to non-ideal tray behavior, heat losses, and limitations in achieving equilibrium, providing insight into the practical operation of continuous distillation columns.

4. Results (15 pts)

VLE graphs made from lab data in order to use the McCabe Thiele method. McCabe Thiele method was compared to Fenske Equation results.

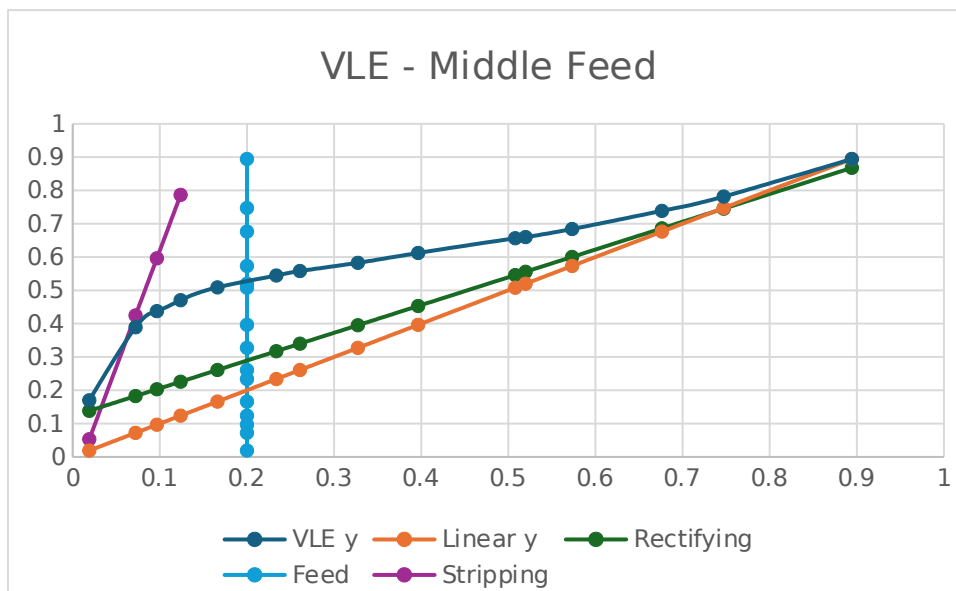


Figure 1: The corresponding VLE lines for experiment B.

Using the McCabe Thiele analysis, 6 stages are counted on the above graph. The efficiency value is therefore 75%. The Fenske equation also resulted in 6 stages.

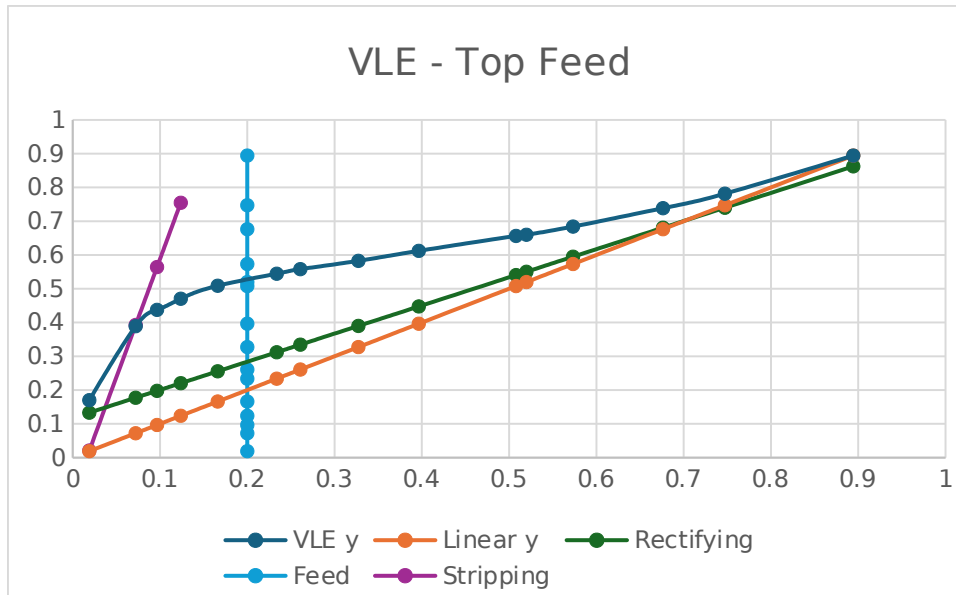


Figure 2: The corresponding VLE lines for experiment C.

Using the McCabe Thiele analysis, 5 stages are counted on the above graph. The efficiency value is therefore 63%. The Fenske equation also resulted in 5 stages.

5. Discussion (15 pts)

While the final temperature data was unable to be collected, the system had a regular increase in temperature until it approached steady state, at which the temperature slowly leveled out. The temperature was recorded when samples were taken, and an interesting note is that for experiment B, where the feed was in the middle, the temperatures of the trays above the feed were cooler than those below the feed. When the feed was at the top of the column in experiment C, the trays were all a consistent temperature. This is likely because the feed stream was cooler than the liquid already within the column.

From the results, the top feed was less efficient at 63% than having the feed in the middle of the column, with an efficiency of 75%. This is because the feed from the middle had to go through more stages and therefore was more thoroughly separated by the time it reached the distillate than when the feed was located at the top of the column. The lower efficiency was reflected by the McCabe Thiele analysis performed and the Fenske equation. However, the McCabe Thiele analysis is not accurate for this column as the CMO is not valid due to the volume present within the different trays not being consistent throughout the column. Additionally, there was approx. a ten degrees Celsius difference in temperature between the reboiler and the condenser, meaning that throughout the column there was also heat loss, and the system was not well insulated. Considering the column is made of glass, this makes sense.

Overall, different tray types and size, and a decrease in the feed flow rate could improve the distillate and bottom concentrations.

6. References (10 pts)

UIC (2024). *Continuous distillation laboratory manual (ChE 382)*.

ChemLibreTexts. (2023). *Distillation and vapor-liquid equilibrium*.
[https://chem.libretexts.org/Bookshelves/Physical and Theoretical Chemistry Textbook Maps](https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps)

Google. (n.d.). WO2000044697A1 - separation of ethanol and water from mixtures thereof by extractive distillation. Google Patents.
<https://patents.google.com/patent/WO2000044697A1/en#>

McCabe, W. L., Smith, J. C., & Harriott, P. (2005). *Unit operations of chemical engineering* (7th ed.). McGraw-Hill.

Seader, J. D., Henley, E. J., & Roper, D. K. (2016). *Separation process principles* (3rd ed.). Wiley.

Towler, G., & Sinnott, R. (2013). *Chemical engineering design: Principles, practice, and economics of plant and process design* (2nd ed.). Elsevier.

Wankat, P. C. (2012). *Separation process engineering: Includes mass transfer analysis* (3rd ed.). Prentice Hall.

7. Appendix I: Data Tabulation/Graphs (10 pts)

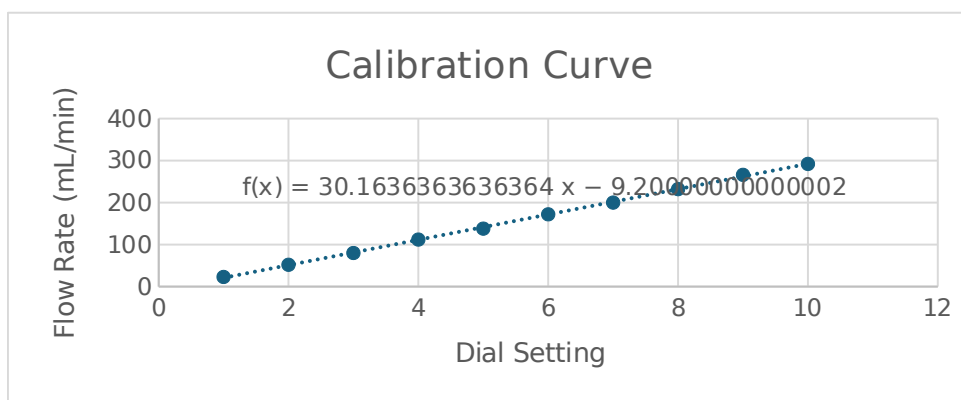


Figure 3: Experiment A calibration curve.

Table 1: Experiment A results for calibration curve.

Setting	Time (min)	Volume (mL)	Flowrate (mL/min)
1	1	23	23
2	0.5	26	52
3	0.5	40	80
4	0.5	56	112
5	0.5	69	138

6	0.5	86	172
7	0.5	100	200
8	0.5	116	232
9	0.5	133	266
10	0.5	146	292

Table 2: Experiment B sample concentration results.

Overhead Sample				Bottoms Sample			
	%v/v	%w/w	mol%		%v/v	%w/w	mol%
1	90.26	87.97	74.09	1	13.80	11.21	4.71
2	89.80	87.42	73.09	2	13.35	10.84	4.54
3	89.90	87.54	73.31	3	10.41	8.40	3.46
Average	89.99	87.64	73.50	Average	12.52	10.15	4.24

Table 3: Experiment C sample concentration results.

Overhead Sample				Bottoms Sample			
	%v/v	%w/w	mol%		%v/v	%w/w	mol%
1	89.00	86.46	71.40	1	16.59	13.56	5.78
2	88.00	85.26	69.35	2	15.22	12.41	5.25
3	88.39	85.73	70.14	3	16.78	13.73	5.86
Average	88.46	85.82	70.30	Average	16.20	13.23	5.63

Table 4: Values used within the VLE plot for Experiment B.**Table 5:** Values used within the VLE plot for Experiment C.

Table 6: Temperature data in Celsius for Experiment B when samples were collected.

Table 6: Temperature data in Celsius for Experiment C when samples were collected.

8. Appendix II: Error Analysis (10 pts)

Both quantitative and qualitative sources of error were considered in evaluating the performance of the continuous distillation column. Experimental uncertainty arose from

measurement limitations, deviations from ideal assumptions, and operational variability within the system.

The primary measured variables were product compositions and temperature profiles. For each operating condition, three replicate samples were collected for both the distillate and bottoms streams, allowing estimation of statistical uncertainty.

For the overhead composition in Experiment B, the three measurements were tightly grouped (e.g., ~90.26, 89.8, 89.9 wt%), giving an average of approximately 89.99 wt% ethanol with a very small spread and standard deviation approximately 0.24 wt%. Similarly, the bottoms composition showed slightly more variation (e.g., 13.8, 13.35, 10.41 wt%), corresponding to a larger standard deviation of approximately 1.8 wt%. This indicated that the distillate composition was measured with higher precision than the bottoms stream.

Using a 95% confidence interval for $n = 3$:

Where $t \approx 4.303$ for $n = 3$, the uncertainty in the overhead composition was small (± 0.6 wt%), while the bottoms stream exhibited significantly higher uncertainty (± 4 -5 wt%). This larger uncertainty in the bottoms composition likely reflected greater sensitivity to flow fluctuations and sampling inconsistencies at lower ethanol concentrations.

Temperature measurements across trays (T1-T9) were highly consistent across replicates, typically varying by less than $\pm 0.3^\circ\text{C}$, indicating that the column had reached a reasonably stable steady state. However, the observed temperature gradient from approximately 79°C at the top to 91°C at the bottom suggested incomplete equilibrium and possible heat losses.

The feed pump calibration also introduced uncertainty. The calibration curve assumed a linear relationship between dial setting and flow rate, but small deviations from linearity and timing/volume measurement errors may have introduced flow rate uncertainty. This directly affected mass balances and operating line construction.

Several systematic and operational factors contributed to deviations from ideal behavior. The McCabe-Thiele method assumes equilibrium on each stage, which is not achieved in practice. The column is also not perfectly insulated, so heat loss reduces vapor generation performance. Despite calibration, the peristaltic pump likely introduced small pulsations in flow rate, affecting steady-state conditions and composition measurements. The densitometer assumes accurate density-composition relationships. Small temperature differences in samples or calibration drift can introduce error, especially at lower ethanol concentrations (bottoms stream). Manual sampling introduced variability in timing and handling. Finally, the analysis assumes $q = 1$, but in practice the feed may not have been perfectly saturated, affecting the feed line and stage calculations.

The relatively small uncertainty in the distillate composition indicated that conclusions regarding top product purity were reliable. However, the larger variability in the bottoms composition introduced greater uncertainty in overall mass balance and efficiency calculations. Additionally, systematic errors such as heat loss and non-ideal stage behavior likely resulted in lower observed separation efficiency than predicted.

Several improvements could reduce experimental uncertainty. Increasing the sampling frequency or number of replicates would improve statistical confidence. Allowing additional time would ensure true steady state conditions before sampling. Using longer collection times during pump calibration would reduce timing error. Improving thermal insulation of the column would minimize heat loss. Using automated or more precise flow control equipment would reduce fluctuations.

9. Appendix III: Sample Calculations (10 pts)

Experiment A was completed by plotting the volume per time to setting data and using the linear trend function within excel to write the formula for the line. The chosen setting was 1.3 on the dial because this was determined to get 30mL/min.

%v/v values were converted to %w/w and then mol% values. Using experiment B overhead sample number one, 90.26 %v/v, the following calculations were done,

And the following calculation was done to get mole%,

Both the Fenske equation and the McCabe Thiele method were used to solve for theoretical stages. An alpha value of 1.8 was used due to literature values, found in the patent "Separation of ethanol and water from mixtures thereof by extractive distillation," stating a range between 1.1 and 2.4 Still using values from experiment B, the following was found

Where $n = 6$.

Efficiency was solved for by dividing the theoretical number of stages by the actual number of stages, or $6/8 \times 100$, resulting in 75%.

Stripping and rectifying equations followed the equations laid out within the theory section.

10. Appendix IV: Individual Team Contributions

NAME: Kendall

	TIME (HOURS)	DESCRIPTION
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Operator (Both Lab Days)	4	Made the water ethanol mixture, ran the controller, collected samples.
Pre-Lab Editing		
Experimental Objective (Prelab)		
Apparatus (prelab)	1	Labeled the apparatus, color coded, and listed the parts in a table
Materials & Supplies (prelab)	0.5	Listed the materials and supplied needed in order to complete this lab
Procedure (prelab)		
Project Safety Evaluation (prelab)		Filled out the safety table, reviewed chemical safety sheet.
Final Lab Editing	0.5	
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	2	Calculated the results from the raw lab data and made the corresponding graphs.
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	1	Made the graphs and made the plots of data
Error Analysis (final lab)		
Sample Calculations (final Lab)	1	
Power Point Presentation	1	
Total	11	

NAME: Lindsey

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Recorded data, constructed calibration curve, took densitometer readings
Pre-Lab Editing	0.5	Formatting and spell check
Experimental Objective (Prelab)	0	

Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	2	Outlined a step by step procedure to follow for both lab days
Project Safety Evaluation (prelab)	0	
Final Lab Editing	0.5	Formatting and spell check
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	3	Used multiple sources to articulate the theory, principles, and governing equations behind the experiment
Results (final lab)	0	
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	0.5	Formatted references into bibliography
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	1	Evaluated results and articulated the cause of errors and how to avoid them in the future
Sample Calculations (final Lab)	0	
Power Point Presentation	1	Translated information to the final presentation
Total	12.5	

NAME: Christian

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Helped operate the apparatus
Pre-Lab Editing	0.5	Finalized labels between tables and figures
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		

Procedure (prelab)		
Project Safety Evaluation (prelab)	1	Wrote about the different safety hazards present within this lab
Final Lab Editing	0.5	Labled the figures and tables and numbered them in the correct order.
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)	3	Connect the results to the theory
Conclusion (final lab)		
References (final and/or prelab)	0.5	Added references to connect the results to the theory
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation		
Total	9.5	

NAME: Srija

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	4	Helped with the lab
Pre-Lab Editing	2	
Experimental Objective (Prelab)	1	Wrote the objective
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	2	
Executive Summary (final lab)	2	Wrote the executive summary
Introduction (Final lab)	2	Wrote the introduction
Theory (Final Lab)		
Results (final lab)		

Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation		
Total	10	